Eric Feng

Principal/Lead Engineer | Unreal Engine Specialist |
Ex-Activision, ex-EA, ex-Wētā FX, ex-ILM

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With over 20 years of dedicated experience in the software industry, I am a passionate programmer whose work has been widely embraced by top-tier game studios, leading visual effects houses, and renowned digital content creation platforms.

WORK EXPERIENCE

Lead Software Engineer
The LEGO Group

12/2024 - Present — *Billund, Denmark*

Achievements/Tasks

- Lead development of the LEGO Digital Brick SDK on Unreal Engine, enabling authentic LEGO brick-building experiences within the UEFN ecosystem.
- Spearhead a cross-functional team to deliver intuitive digital building UX, high-fidelity brick physics, character animation systems, and performance-optimized tooling for creators.

Principle Gameplay Engineer
Self-Employed

06/2022 - Present — *Remote*

Achievements/Tasks

- Independent Game Development Consultant specializing in gameplay systems and animation.
- Collaborate closely with design and animation teams to architect and implement advanced gameplay mechanics, with a focus on character traversal, 3C (Character, Camera, Controls), and AI behavior.
- Deliver high-impact solutions that elevate player experience and technical fidelity across diverse projects.

Expert Gameplay Engineer
Sledgehammer Games, Activision

11/2021 - 05/2022 — *Melbourne, Australia*

Achievements/Tasks

- Research and implement gameplay mechanics for up-coming Call Of Duty games.
- Optimize loot pickup and destruction system with ECS framework.

Software Engineer
Frostbite - EA

03/2020 - 10/2021 — *Melbourne, Australia*

Achievements/Tasks

- Main developer at the core team of Entity-Component-System (ECS) framework in Frostbite Engine.
- Collaborate in features development with different game teams around the globe.

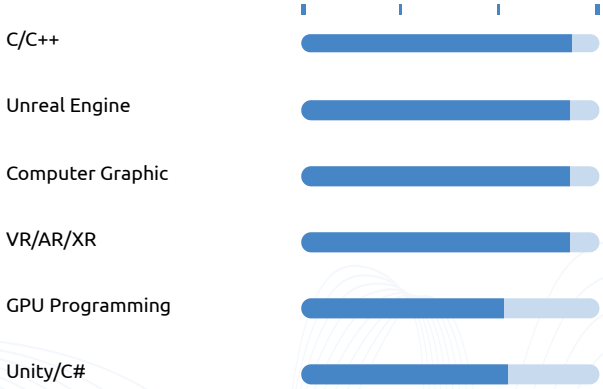
Senior Software Developer
Wētā FX

08/2013 - 03/2020 — *Wellington, New Zealand*

Achievements

- Designed and developed multiple award-winning real-time interactive VR/ AR/ XR applications in Unreal Engine and Unity with a focus on gameplay, character animation, rendering and performance profiling.
- Main developer for an academy award-winning scene description system (Atlas) focusing on low-level API design, animation, evaluation system and real-time rendering for the proprietary scene data.

TECHNICAL SKILLS



PROJECTS

Lego Brick Editor in Unreal Engine / UEFN (12/2024 - 04/2026)

- A collaboration project with Epic Games. It is used in LEGO Fortnite and UEFN.
- Lead Software Engineer. Have been working on user experience of digital brick building, animations, physics and optimization.
- Support both internal and external vendors for using Atom framework in UE5 projects.

Exodus (04/2023 - 12/2024)

- Mostly focused on player traversal gameplay mechanics and gameplay animation implementations.
- Product owner and main contributor for swing, grapple, balance beam and camera transition systems.
- Ensured the gameplay systems work well with GAS, companion systems and NPC interactions.

Call of Duty: Modern Warfare III (12/2021 - 05/2022)

- Created a new Point-Of-Interest (POI) for NPCs.
- Adopted ECS system for gameplay system optimization.
- Underwater gameplay animation setup.

Towerborne (11/2020 - 12/2021)

- Co-development consultant on Stoic's action-RPG, built in Unreal Engine.
- Ported the game to Xbox from Windows.
- Set up the Epic Online Services subsystem integrated with PlayFab.
- Performance tuning for character spawning, garbage collection, and game load times to hit the framerate target on all supported platforms

Frostbite Engine (03/2020 - 10/2021)

- Design and develop foundation modules for Entity-Component-System (ECS) in Frostbite engine.
- Collaborate with tech artists to define new workflows for FX, Lighting and Animation.

Childish Gambino's Pharos (09/2018 - 12/2018)

- 2019 VES Award for Outstanding Visual Effects in a Special Venue Project.
- Main developer for Dome projection VR preview, gameplay and interactive graphics for the concert.

R&D Engineer
Industrial Light & Magic

09/2011 - 07/2013 *Singapore*

Achievements

- Designed and developed academy Sci-tech award-winning ILM's proprietary procedural rigging system, Block Party 2.
- Developed HoudiniBatch, a render farm manager for Houdini allowing FX artists to maximize the efficiency of the academy award-winning ObaQ render queue management system.

Motion Capture Pipeline Technical Director
Framestore

09/2010 - 09/2011 *London, UK*

Achievements

- Built up the whole motion capture data pipeline and production tools from scratch, including data repository management, in-house Motionbuilder real-time control panel design/develop and Vicon mocap processing pipeline.
- Worked on stage to provide technical supports for multiple shows including Gravity (2013), Wrath of Titan (2012) and etc.

Software Engineer
Autodesk

12/2007 - 08/2010 *Shanghai, China*

Achievements

- Designed, developed and maintained Autodesk FBX Library (includes SDK and plug-ins) , Autodesk Material Library and Autodesk Mudbox
- Collaborated with multiple Autodesk product teams in different countries on FBX pipeline
- Assisted high profile clients in resolving technical issues.

EDUCATION

Computer Science (09/2002 - 07/2006)
Shanghai Jiao Tong University
Shanghai, China. GPA 3.6/4